



Applying the four operations to fractions

Teacher resource

Question 1

How much is three-fifths of \$400?

\$80
☐

\$120
☐

\$160
☐

\$240
☐

\$320
☐

Answer: D

Skill: Students calculate a fraction of a quantity.

Additional questions

1. Would three quarters of \$400 be more or less than \$240?
2. How can you use the fact that three fifths of \$400 is \$240 to find three fifteenths of \$400?
3. If \$240 is three-fifths of \$400, what is three-fifths of \$1200?

Question 2

What number, when subtracted from $3\frac{2}{3}$, gives $1\frac{5}{6}$?

$2\frac{1}{2}$
☐

$1\frac{5}{6}$
☐

$3\frac{5}{6}$
☐

$2\frac{1}{6}$
☐

Answer: B

Skill: Students add and subtract fractions

Background information

Students should be encouraged to use informal strategies, rather than formal written procedures, when calculating with fractions that are easily visualised or have well known equivalences.

For example, $2\frac{1}{2} + 2\frac{3}{4}$ can be partitioned as $2\frac{1}{2} + 2\frac{1}{2} + \frac{1}{4}$ to make this calculation

relatively simple. Such a strategy can be used for mental calculation or, if needed, recorded in writing as an informal written calculation. In contrast, the use of a formal written calculation, involving conversion to equivalent fractions with the same denominators, to carry out this calculation would increase both the complexity of the task and the opportunities for error.

Additional questions

1. What number subtracted from $5\frac{1}{4}$ gives $2\frac{1}{2}$?
2. What is the value of $5\frac{1}{4} + 3\frac{1}{2}$?
3. What is the sum of $3\frac{1}{2}$ and $2\frac{1}{3}$?

Question 3

$$42 \times \Delta = 30$$

Which fraction is Δ equal to?

$$\frac{6}{7}$$

☐

$$\frac{5}{6}$$

☐

$$\frac{5}{7}$$

☐

$$\frac{1}{6}$$

☐

Answer: C

Skill: Students connect multiplication and division with fractions

Background information

Some students may solve this question through trial and error, testing each of the given fractions in turn. This strategy might not be useful, however, if the numbers were more complex. Students may need support to use the inverse relationship between multiplication and division to rearrange and solve the equation. While students may use this relationship to solve equations when the unknown is a whole number, they may need support to interpret the answer when the unknown is a fraction.

Additional questions

What number is represented by the Δ in these equations?

1. $42 \times \Delta = 21$
2. $\Delta \times 42 = 21$
3. $21 \div \Delta = 42$
4. $1/2 \div \Delta = 42$
5. $\Delta \div 1/2 = 21$

Question 4

A farmer had a full container of pellets to feed his chickens.
He used one-third of the pellets on the first day.
On the second day he used one-quarter of the remaining pellets.
There were 15 kilograms of pellets left in the container after the second day.

What was the weight of pellets in the full container?

kilograms

Answer: 30 kilograms

Skill: Students calculate fractions of fractions.

Additional questions

Determine the value of each of the following:

1. one quarter of one half
2. one third of one quarter
3. two thirds of one quarter
4. two thirds of two thirds

Student activity: Applying the four operations to fractions

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